Selective Disclosure of Agent Capabilities (sd-jwt)

IETF 125 - AI Protocols Side Meeting

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Privacy Problem: Agent Cards Expose Everything

SD-CARD

Traditional Agent Card

Name: Enterprise AI Assistant

Skills: code-review, data-analysis, hr-lookup, salary-calculator, admin-functions

Endpoints: public-api, internal-api, debug-api

Provider: Acme Corp - Internal IT

Security: oauth, api-key, admin-token

Partner Agent | Public Query | **Attacker**

ALL see SAME information

PROBLEM	RISK
Information Leakage	All capabilities exposed to any entity
Weak Authentication	No cryptographic proof of identity
Static Disclosure	Same info shown to everyone
Cross-Context Linkability	Identical presentations enable tracking

Without SD-CARD
Full card to everyone = privacy leak

With SD-CARD
Tailored per context = minimal disclosure

SD-Card: Selective Disclosure for Agents

SD-CARD

SD-Card = Agent Card + SD-JWT (Selective Disclosure JWT per RFC 9901)

Same Agent, Different Views

Public Context

Name: AI Agent

Skills: text-gen

External Users

Internal Context

Name: AI Agent

Skills: text-gen, code-review,

Endpoints: public, internal

Internal Teams

Diagnostic Context

Name: AI Agent

Skills: [ALL]

Endpoints: [ALL]

Security: [ALL]

IT Operation

Key Benefits

Privacy
Minimal disclosure principle

Security
Cryptographic key binding

Unlinkable
Unique salts prevent tracking

SD-Card: Selective Disclosure for Agents

SD-CARD

SD-JWT Structure

```
<Issuer-Signed JWT>~<Disclosure 1>~<Disclosure 2>~<KB-JWT>

+------------------+  +------------------+
| BASE CLAIMS      |  | DISCLOSURES      |
| (Always Shown)   |  | (Optional)       |
+------------------+  +------------------+
| - iss (issuer)   |  | - skills         |
| - sub (agent id) |  | - capabilities   |
| - exp (expiry)   |  | - endpoints      |
| - cnf (pubkey)   |  | - provider       |
| - _sd (digests)  |  | [salt,key,value] |
+------------------+  +------------------+

+------------------+
| KEY BINDING JWT  |
| (Fresh per use)  |
+------------------+
| - iat (time)     |
| - aud (verifier) |
| - sd_hash        |
| Signed by Agent  |
+------------------+
```

Questions ?

draft-nandakumar-agent-sd-jwt-02
https://datatracker.ietf.org/doc/draft-nandakumar-agent-sd-jwt/

Thank you!